21 Electric Fields

Static electricity can often be a nuisance in everyday life.
Have you ever had an electric shock when getting out of a car?



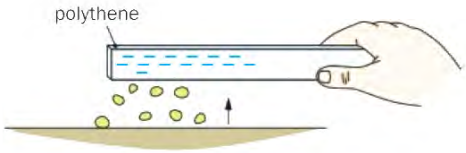
Lightning is a spectacular display of static electricity.
In 1752, in a famous but highly dangerous experiment, Benjamin Franklin flew a kite during a thunderstorm and proved that lightning is a form of electricity.
Some people were killed as they tried to repeat Franklin's work!

You have already studied moving charges or electric current in Chapter 16.
In ***static*** electricity, the electric charges are either at rest or only flow for a very short time.

Electrostatics has many useful applications.
These include the electrostatic spraying of paints and powders and removing smoke particles from power station chimneys.

In this chapter you will learn:

- how to calculate the electric force between two charges,
- about electric fields,
- about electric field strength and electric potential.

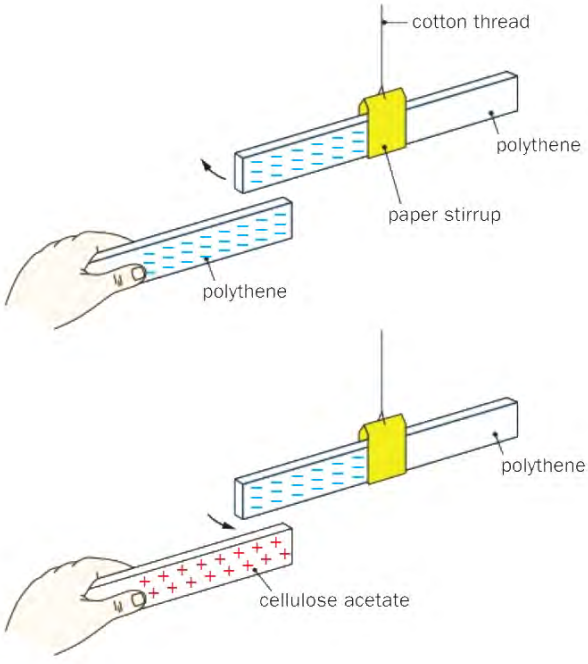


The diagrams show some simple experiments in electrostatics that you may have done before.

Do you remember the following facts?

- Objects are usually uncharged and so are electrically neutral.
- Objects can become charged by friction, when one material is rubbed against another material. This is discussed on the opposite page.
- A charged object can attract an uncharged object, for example, small pieces of paper.
- There are two types of electric charge. We call them positive and negative.
- A rubbed polythene strip gains a negative charge. A cellulose acetate strip gains a positive charge.
- When two charged objects are brought together, we find that:

  Like electric charges repel.
  Opposite electric charges attract.

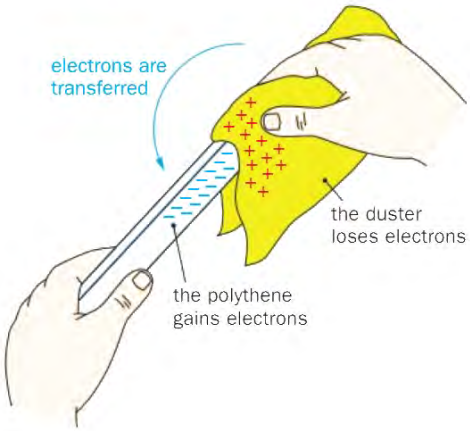


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How do objects become electrically charged?

Normally, objects are uncharged because each atom has the same number of positive protons as negative electrons. Charging occurs when ***electrons are transferred*** from one material to another.

Friction can transfer electrons, as shown in the diagram:



The polythene has gained extra electrons and so it is negatively charged. The duster is left short of electrons. What is its charge?

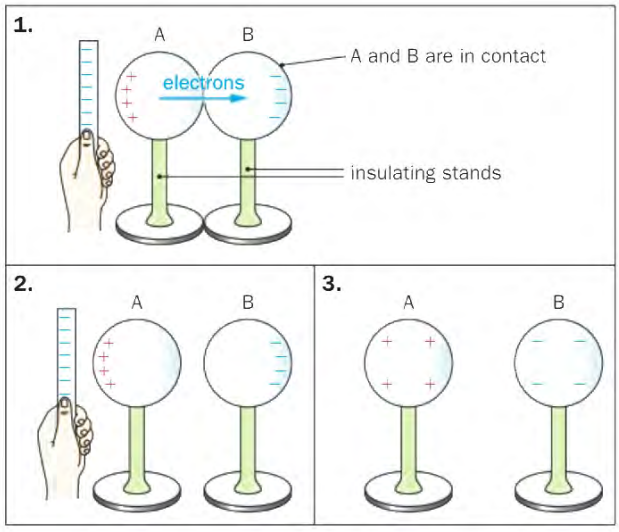
Polythene (like all insulating materials) has few free electrons (see Chapter 16) and so the negative charge does not flow away from the rubbed region – it is a ***static charge***.

Can you put a metal rod in your hand and charge it by rubbing? Any charge transferred to the rod will flow along the rod and through your body to Earth, because your body and the Earth are poor insulators of electricity.
A conductor can only be charged if it is insulated from Earth.

Charging by electrostatic induction

A charged conductor will share its charge with an uncharged conductor that is placed in contact with it. Electrons can flow between the two conductors because they are touching.

A charged strip can also be used to ***induce*** a charge in an uncharged conductor ***without*** touching it.
The diagrams show how two conducting spheres can be oppositely charged by **induction**:



1. When a negative strip is brought close to 2 metal spheres, electrons from sphere A are repelled to sphere B.
2. The spheres are separated with the strip still held nearby.
3. Then the strip is removed. The free electrons in each sphere now spread out, so that the charge is distributed evenly over the surface of each sphere.

Why does a balloon, charged by rubbing, stick to a wall?
The negative charge on the balloon repels some of the electrons in the wall away from the surface. This leaves the wall surface positively charged, and so the balloon is attracted to the wall.

Measuring charge

One way of measuring charge is to use a **coulombmeter**:
Some of the charge on the polythene strip is transferred to the coulombmeter when the strip is scraped across its metal cap. The coulombmeter then measures the charge placed on its cap.

Why is this coulombmeter giving a negative reading?
This confirms that the polythene strip is negatively charged.

The size of this charge is 32 nC or 32×10^{-9} coulombs.
Remember, the coulomb is a very large unit (see page 228).
In fact, the coulombmeter has been given 200 billion electrons from the polythene strip!



Charging a coulombmeter.

Forces between charges



What affects the force that charged objects exert on each other? You can investigate this using the apparatus in the photograph:

Place a charged polythene strip on top of an insulating material on an electronic balance and set the balance to read zero. What happens as you bring up a second charged polythene strip?

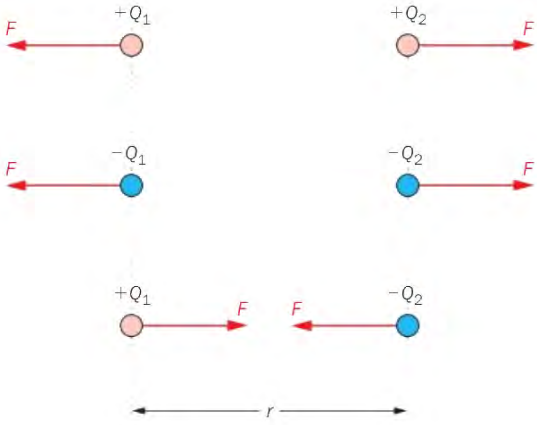
The strip in your hand repels the strip on the balance and this downwards force causes the balance reading to increase.

Does the force increase as the strips are brought closer together?
Does the force increase if you increase the charge on each strip?

Coulomb's Law

In 1785, the French physicist Charles Coulomb carried out a series of experiments and discovered that the force F between two charges was:

- directly proportional to each of the charges Q_1 and Q_2,
- inversely proportional to the ***square*** of their separation r.



Coulomb's Law is written:

$$F \propto \frac{Q_1 Q_2}{r^2} \quad \text{or} \quad \boxed{F = \frac{k\,Q_1\,Q_2}{r^2}}$$

where: F is the force, in newtons
Q_1 and Q_2 are the charges, in coulombs
r is the distance apart, in metres
k is the constant of proportionality

Strictly, Coulomb's equation applies only to point charges, but a sphere of charge behaves as if all its charge is concentrated at its centre (providing it is not placed too close to another charge).

The value of the constant k



When the two charges are separated by a vacuum, the value of the constant k is 9.0×10^9 N m^2 C^{-2}. When they are separated by air, it is effectively the same.

In fact, this is the largest value that k can have because the size of the force is always reduced when an insulating material, other than air, separates the two charges.

In a vacuum (or air) the value of k can be written as: $k = \dfrac{1}{4\pi\varepsilon_0}$

where ε_0 is a constant called the **permittivity of free space**.
Its value is 8.85×10^{-12} F m^{-1} (The farad, F, is explained in Chapter 22.)

So $k = \dfrac{1}{4\pi\varepsilon_0} = \dfrac{1}{4\pi \times 8.85 \times 10^{-12}\ \text{F m}^{-1}} = 9.0 \times 10^9$ N m^2 C^{-2}

Every insulating material has a permittivity ε which is greater than ε_0.

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Another inverse-square law

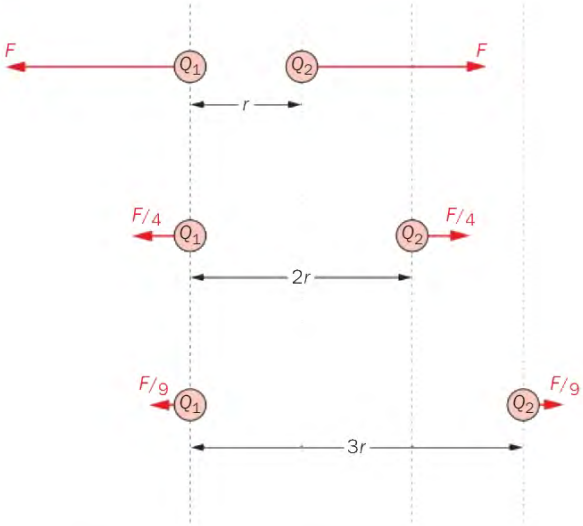
Can you see that the two equations below are very similar?

Coulomb's law:

$$F = \frac{k\,Q_1\,Q_2}{r^2}$$

Newton's law of gravitation (page 86):

$$F = \frac{G\,m_1\,m_2}{r^2}$$



Coulomb's law is similar to Newton's law, except that for gravity the force can only attract, whereas the force between charges can attract ***or*** repel. The diagram shows 2 charges repelling:

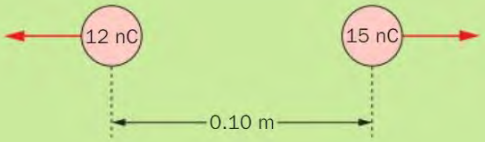
Like Newton's law, Coulomb's law is an inverse-square law. This means that if we double the distance between the charges, the force decreases to $(\frac{1}{2})^2$ or $\frac{1}{4}$ of its previous value.

Look at the diagram:
Why does the force drop to $^F/_9$ when the separation is tripled?
The force decreases to $(^1/_3)^2$ or $^1/_9$ of its previous value.

We can make other comparisons between these two laws.
Can you find the similarities and the differences in the table?

Newton's law of gravitation	Coulomb's law of charge
The force is 'felt' by objects with mass.	The force is 'felt' by objects with charge.
The force is proportional to the size of the masses.	The force is proportional to the size of the charges.
The force is inversely proportional to the square of the separation of the masses.	The force is inversely proportional to the square of the separation of the charges.
There is a gravitational field round a mass.	There is an electric field round a charge (see page 292).
There is only one type of mass.	There are two types of charge – positive and negative.
The force is always attractive.	The force can be attractive or repulsive.
The constant of proportionality is G. G always has the value 6.67×10^{-11} N m^{-2} kg^{-2}.	The constant is k, and its value depends on the material. In air, $k = {}^1/_{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^{-2} C^{-2}.
The gravitational force is very weak, unless one of the masses is large.	The electric force is very strong, but not normally noticed as the charges cancel out within a neutral atom.

Example 1
Two small conducting spheres are placed 0.10 m apart in air, and are given positive charges of 12 nC and 15 nC, as shown:
Calculate the size of the repulsive force between them.
(The permittivity ε_0 of air is 8.85×10^{-12} F m^{-1}.)



$$F = \frac{k\,Q_1\,Q_2}{r^2} \quad \text{where } k = \frac{1}{4\pi\varepsilon_0}$$

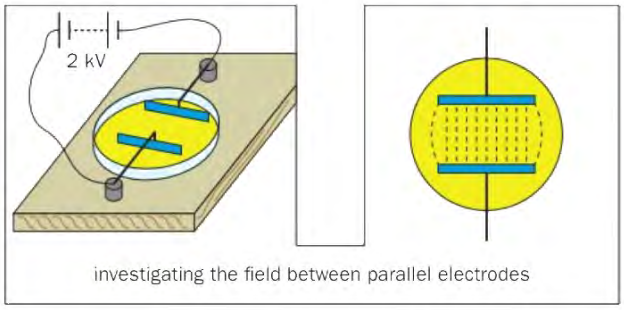
$$\therefore\ F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2} = \frac{12 \times 10^{-9}\ \text{C} \times 15 \times 10^{-9}\ \text{C}}{4\pi \times 8.85 \times 10^{-12}\ \text{F m}^{-1} \times (0.10\ \text{m})^2} = 1.6 \times 10^{-4}\ \text{N}\ \ (2\ \text{s.f.})$$

Electric fields

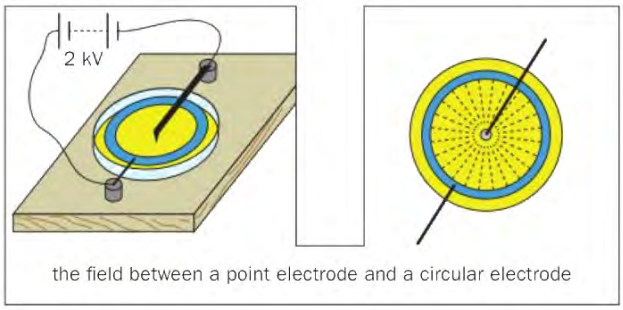


The charged polythene strip is attracting Phiz's hair towards it.
We say that there is an **electric field** around the strip.
An electric field is a region where a charge 'feels' a force, just as a gravitational field is a region where a mass 'feels' a force.

Is it possible to see the shape of an electric field?
One way is to use semolina grains sprinkled on castor oil.
Two electrodes in the oil produce the electric field, as shown.
The semolina grains line up along the electric field lines:



investigating the field between parallel electrodes



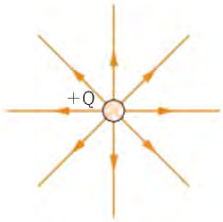
the field between a point electrode and a circular electrode

The arrows on the field lines tell us the ***direction*** of the field.
This is defined as the direction of the force on a ***positive*** charge and so the arrows on the lines go from positive to negative.

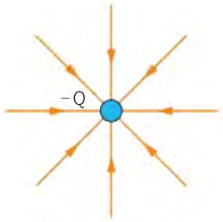
The spacing of the lines tells us about the ***strength*** of the field.
The more closely packed the field lines, the stronger the field.

Look at the electric field caused by the parallel electrodes.
The field lines are parallel to each other and equally spaced.
This field is ***uniform***. It has the same strength at all points.

The field of the point electrode is called a ***radial*** field.
The diagrams show the radial field of an isolated point charge.
How can you tell that the field is strongest close to the charge?



this radial field is repulsive



this radial field is attractive

Electric field strength, *E*

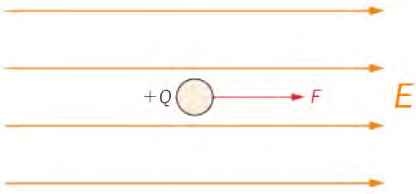
On page 88 we defined gravitational field strength g as the force acting per kilogram of mass.
Electric field strength E is defined in a similar way:

The electric field strength at a point is the force per coulomb exerted on a positive charge placed at that point in the field.

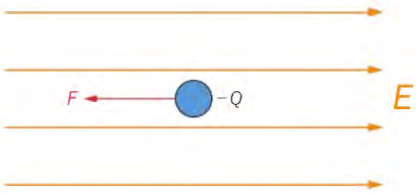
The diagram shows a charge $+Q$ placed in a field of strength E:
If E is the force per coulomb (in newtons per coulomb), then:

Electric field strength, E (N C^{-1}) $=$ **force, F** (N) / **charge, Q** (C) or $E = \frac{F}{Q}$

It follows that: $F = QE$



The charge +Q 'feels' a force F in the same direction as E.



The force on the negative charge −Q is in the opposite direction to E.

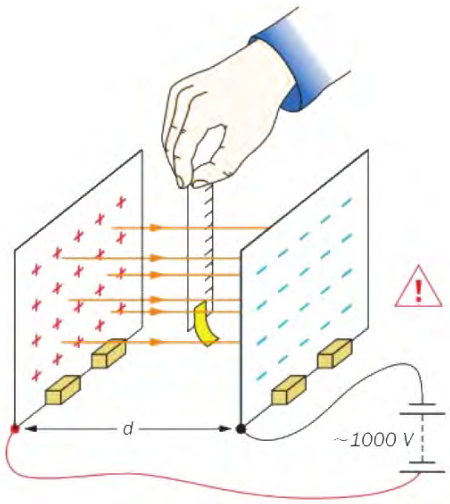
- Electric field strength E is a vector quantity. The direction of the field is the direction of the force on a positive charge.
- The unit here for E is N C^{-1}, but an alternative unit is volts per metre (V m^{-1}). You will see why on the next page.

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The uniform electric field

In the diagram, there is a p.d. of about 1000 V across the parallel metal plates, so that they become charged as shown:

A ***uniform*** electric field is produced between the plates.



You can test the electric field using a home-made **electroscope**. This consists of a strip of metal foil attached to a plastic ruler. You charge the metal foil and hold the electroscope in the field. The stronger the field, the greater the force on the charged foil, and the more it deflects from the vertical position.

How can you make the field stronger? You can:

- increase the p.d. across the plates, or
- move the plates closer together.

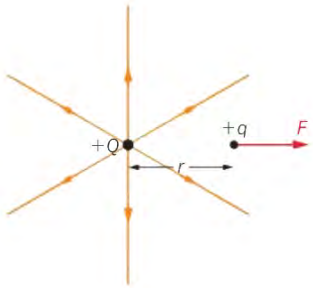
In fact, the field between the plates can be calculated by:

Electric field strength, E (V m^{-1}) = p.d. across the plates, V (V) / separation of plates d (m) or $E = \frac{V}{d}$

Note that the unit here for E is V m^{-1}. (1 V m^{-1} = 1 N C^{-1})

The radial electric field

What is the electric field strength at any point in a ***radial*** field?
In the diagram, a point charge $+Q$ produces the radial field, and a ***small*** positive test charge $+q$ is placed in the field as shown:



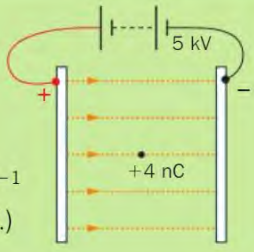
Using Coulomb's Law the force on the test charge $+q$ is given by: $F = \frac{kQq}{r^2}$ where $k = \frac{1}{4\pi\varepsilon_0}$

The electric field strength is the force per unit charge. Therefore: $E = \frac{F}{q} = \frac{kQ\cancel{q}}{\cancel{q}r^2}$

So the field strength E a distance r away from a point charge $+Q$ is: $\boldsymbol{E = \frac{kQ}{r^2}}$

The field strength E a distance r away from a negative point charge $-Q$ is: $\boldsymbol{E = -\frac{kQ}{r^2}}$
The − sign tells us the field is inwards.

Example 2
Two plates are placed 10 cm apart and connected to a 5.0 kV supply.
Calculate: a) the strength of the uniform field between the plates,
b) the force on a +4.0 nC charge placed in the field.



a) $\boldsymbol{E = \frac{V}{d}}$

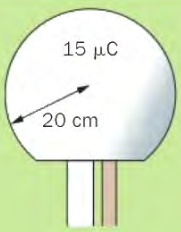
$E = \frac{5.0 \times 10^3 \text{ V}}{0.10 \text{ m}} = 5.0 \times 10^4 \text{ V m}^{-1}$

b) $\boldsymbol{F = QE}$

$F = 4.0 \times 10^{-9} \text{ C} \times 5.0 \times 10^4 \text{ N C}^{-1}$

$F = 2.0 \times 10^{-4}$ N, to the right (2 s.f.)

Example 3
The charge on the dome of a Van de Graaff generator is 15 μC.
The dome radius is 20 cm. What is the field strength at its surface?
Remember, a sphere acts as if all its charge is at its centre.



$\boldsymbol{E = \frac{kQ}{r^2}}$ where $k = \frac{1}{4\pi\varepsilon_0}$ and ε_0 is 8.85×10^{-12} F m^{-1}

$E = \frac{Q}{4\pi\varepsilon_0 r^2} = \frac{15 \times 10^{-6} \text{ C}}{4\pi \times 8.85 \times 10^{-12} \text{ F m}^{-1} \times (0.20 \text{ m})^2} = 3.4 \times 10^6 \text{ V m}^{-1}$ (2 s.f.)

Electric potential

If Phiz wants to move his charge closer to the charged sphere he has to push against the repulsive force:
Phiz ***does work*** and his charge ***gains*** electric potential energy.

If Phiz lets go of the charge it will move away from the sphere, losing electric potential energy, but gaining kinetic energy.

When you move a charge in an electric field, its potential energy changes. This is like moving a mass in a gravitational field.

Gravitational potential was discussed in Chapter 8.
Electric potential is a similar quantity:

The **electric potential *V*** at any point in an electric field is the ***potential energy that each coulomb of positive charge would have if placed at that point in the field.***

The unit for electric potential is the joule per coulomb (J C^{-1}), or the **volt** (V). Like gravitational potential it is a ***scalar*** quantity.



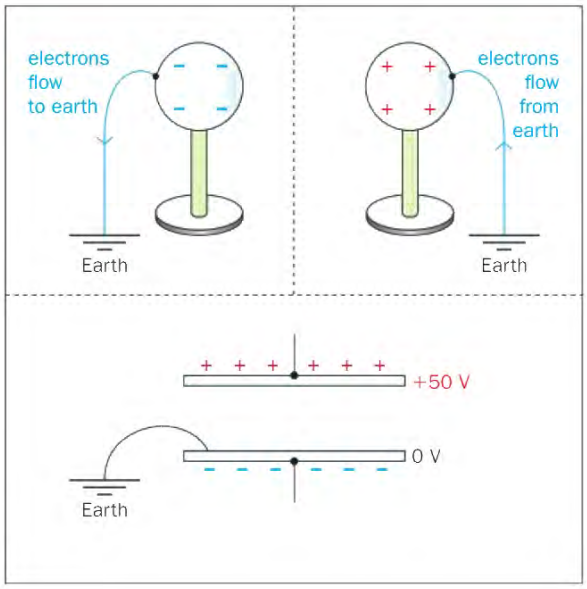
Earthing

In electrostatics, all the points on a conducting surface are at the same potential. If not, electrons would flow until the potential of all of the conductor becomes the same.

Where is the zero of electric potential?
Strictly, electric potential is defined to be zero at infinity, just like it was for gravitational potential on page 92.
In practice, it is convenient to say that Earth is at zero potential.
Why must a conductor connected to Earth be at zero potential?

These spheres lose their charge when they are earthed:
Can you see which way the electrons flow in each case?

A conductor can retain a charge even when it is earthed!
Look at the parallel plates:
The lower plate is at zero potential (earthed), but even so it has a negative charge which is attracted by the nearby positive plate.

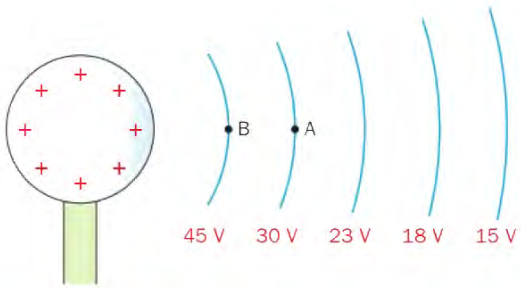


Potential difference

We often need to know the difference in potential between two points in an electric field:

The potential difference or p.d. is the energy transferred when one coulomb of charge passes from one point to the other point.

The diagram shows some values of the electric potential at points in the electric field of a positively charged sphere:
What is the p.d. between points A and B in the diagram?



When one coulomb moves from A to B it gains 15 J of energy.
If 2 C move from A to B then 30 J of energy is transferred. In fact:

Energy transferred, ΔW (joules) **= charge, Q** (coulombs) **× p.d., ΔV** (volts)

or

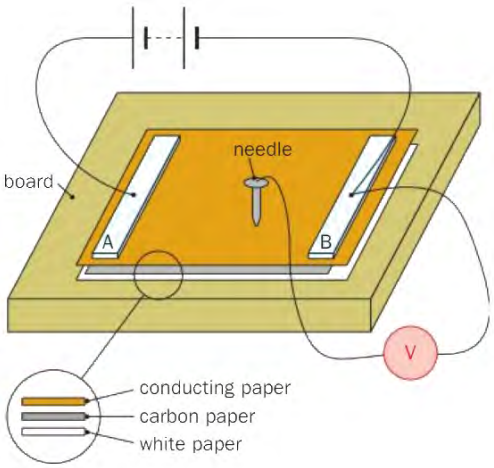
$$\Delta W = Q \times \Delta V$$

Do you recognise this equation from Chapter 16?
P.d. has the same meaning in static and current electricity.

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Equipotential surfaces

All the points that have the ***same*** potential in an electric field are said to lie on an **equipotential** surface.
The diagram shows how you can investigate equipotentials:

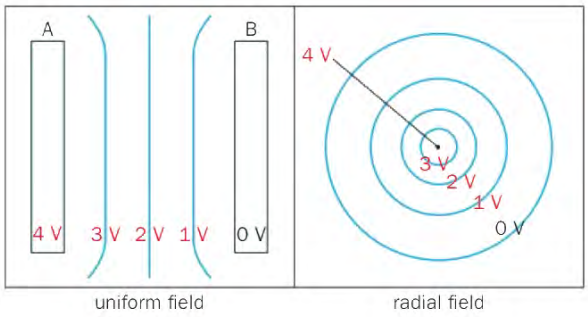


The 2 electrodes A and B are clamped to the conducting paper. You can trace the equipotential lines using the voltmeter. It measures the p.d. between the contact needle and electrode B.

You can place the needle on electrode A and adjust the d.c. supply until the voltmeter reads 4.0 V.
Then move the needle across the paper, so that you draw a line through all the points where the voltmeter reads 3.0 V. Then draw lines for points where the potential is 2.0 V and 1.0 V.

You can draw the equipotentials in a radial field by repeating this experiment using a point electrode and circular electrode.

The diagrams show the results:



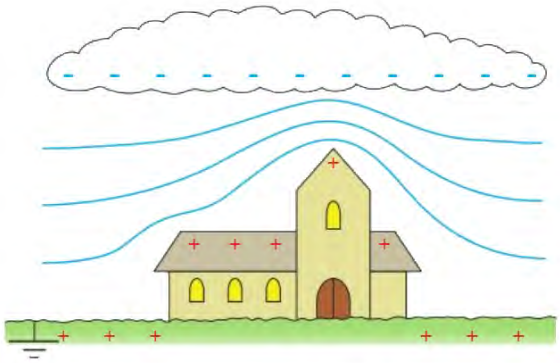
Look at the equipotentials between the **parallel** plates:
They are evenly spaced because the electric field is ***uniform***.

In a **radial** field the equipotentials are concentric spheres:
Why are they closer together near the point charge?
Imagine moving a small positive test charge in this field.
As you get closer to the point charge the field gets stronger, the repulsive force increases and you have to do more and more work to move the test charge the same distance.

When a charge moves ***along*** an equipotential, ***no work*** is done because its potential energy does not change.
This means that the equipotentials must cross electric field lines at 90°, just like in gravitational fields (see page 92).

Lightning

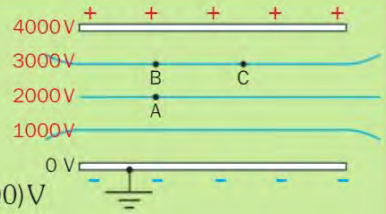
An electric field of about 3×10^6 V m^{-1} will ionise dry air.
This means that there must be a p.d. of around 3 million volts across each metre of air for it to conduct electricity.



Look at the equipotentials between the cloud and the church:
Can you see how the pointed spire distorts these lines?
The field is strong where the equipotentials are close together.

Why is the lightning most likely to strike the spire?
The field around the spire is strong enough to ionise the air, so that charge can flow between the cloud and the spire.

Example 4
The diagram shows the equipotentials between two plates.
How much energy is transferred when a charge of 2.0 nC moves a) from point A to B, b) from B to C?



$\Delta W = Q\Delta V$ where V is the p.d. between the points

a) $\Delta W = 2.0 \times 10^{-9}\text{C} \times (3000 - 2000)\text{V}$
$= 2.0 \times 10^{-9}\text{C} \times 1000\text{V}$
$= 2.0 \times 10^{-6}\text{J}$ (2 s.f.)

b) $\Delta W = 2.0 \times 10^{-9}\text{C} \times (3000 - 3000)\text{V}$
$= 2.0 \times 10^{-9}\text{C} \times 0\text{V}$
$= 0\text{J}$ (on the same equipotential)

Electric potential in a radial field

The electric potential V at a point in the field of a point charge depends on the charge Q and the distance r from the point charge.

For a positive point charge: $\boldsymbol{V = \frac{kQ}{r}}$ and for a negative point charge: $\boldsymbol{V = -\frac{kQ}{r}}$ where $k = \frac{1}{4\pi\varepsilon_0}$

(This value of k applies when the charge is in a vacuum or in air; see page 290.)

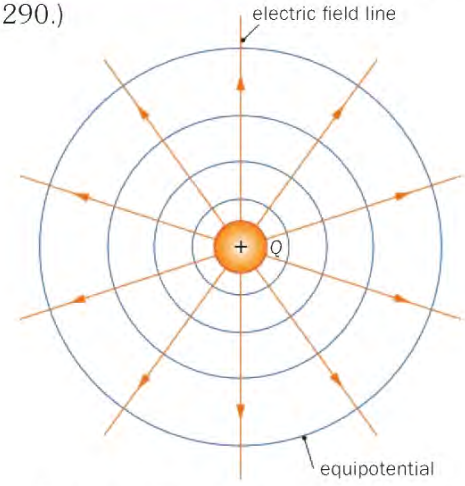
Can you see that as r becomes larger, the potential gets smaller?
The absolute electric potential at a point in an electric field is defined as:

the work done per unit positive charge, when the charge is moved from infinity to that point.

So $V = 0$ at infinity (as we said on page 294).

Why is the potential positive for charge $+Q$ and negative for $-Q$?

Suppose you move a small positive test charge ***towards*** the charge $+Q$. You have to ***do work against*** the repulsive force, and so the test charge gains energy.
But for the charge $-Q$ we do work against the attractive force as the positive test charge moves further ***away***. V must have negative values so the energy of the test charge can increase and reach zero at infinity.



The electric field of a positive point charge.

Linking electric potential V and field strength E

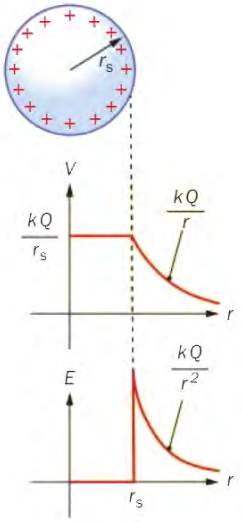
When a charge $+Q$ is placed on an isolated conducting sphere, the charge spreads uniformly over the surface of the sphere.
Here the radius of the sphere is r_s.

The upper graph shows how the electric **potential** V varies with the distance r from the centre of the sphere:
All of the conducting sphere has the same potential V, where $\boldsymbol{V = \frac{kQ}{r_s}}$ (the same value as if all the charge were at its centre).
Notice how, beyond r_s, the potential V decreases as $1/r$.

The lower graph shows how the electric **field strength** E varies with the distance r from the centre of the sphere:
At any point ***outside*** the sphere, E decreases as $1/r^2$ and has the same value as if all the charge is at the centre of the sphere.

In fact, the negative gradient of the V graph at any point gives the value of E, the electric field strength, at that distance: $\boldsymbol{E = -\frac{\Delta V}{\Delta r}}$

For $r \leq r_s$, the gradient of the V graph is zero, and so inside the conducting sphere the electric field strength E is zero.



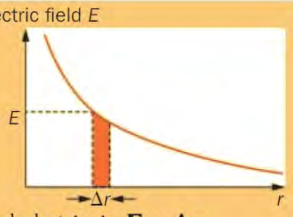
Example 5
An isolated conducting sphere of radius 12 cm is given a charge of 0.16 nC.
What is the electric potential of the sphere?

$$\boldsymbol{V = \frac{Q}{4\pi\varepsilon_0 r}} = \frac{0.16 \times 10^{-9}\ \text{C}}{4\pi \times 8.85 \times 10^{-12}\ \text{F m}^{-1} \times 0.12\ \text{m}}$$

$= \underline{12\ \text{V}}$ (2 s.f.)

$\boldsymbol{E = -\frac{\Delta V}{\Delta r}}$

$\therefore \boldsymbol{\Delta V = E\Delta r}$



The area of the shaded strip is $\boldsymbol{E \times \Delta r}$.
So adding up the total area under the curve would give us the total change in potential, $\boldsymbol{\Delta V}$.

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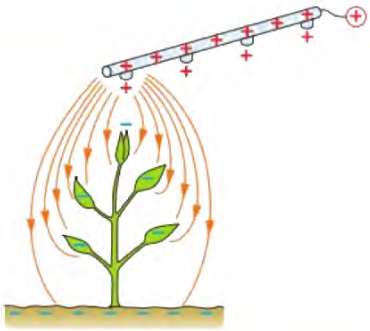
Physics at work: Using static electricity

Electrostatic crop spraying

Most crop pests live on the underside of leaves. Electrostatics can be used to target these 'hard to get at' places.
The boom of the spray is connected to a high-voltage supply.
This produces positively charged pesticide droplets, and also induces an opposite charge in the plants and in the ground.

The droplets repel each other to form an even cloud of spray, and are attracted to all parts of the oppositely charged plant.
Can you see how the spray follows the electric field lines and reaches the underside of the leaves?

Bicycles and cars are painted using a similar technique.



The photocopier

The diagrams show four of the stages in Xerox photocopying.

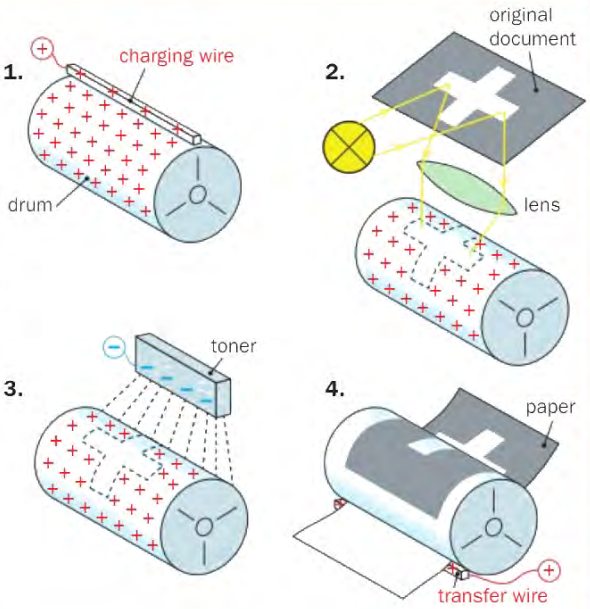
In stage **1**, a high-voltage wire charges the insulating surface of the drum. The drum is now positively charged.
An intense light will allow charge to escape from this drum.

In stage **2**, intense light is flashed across the original document.
The reflected light is then projected on to the drum surface.
Which parts of the drum remain charged?
The black parts of the original document do not reflect the light and so the corresponding regions of the drum remain charged.
The drum surface now holds an image of the original.

In stage **3**, negatively charged toner particles are attracted to the parts of the drum which are still positively charged.

In stage **4**, a sheet of paper is passed over the drum surface.
There is a positively charged transfer wire below the drum.
This attracts the toner from the drum on to the paper.

Finally, the paper is passed through heated rollers, which melt the toner and fuse it to the paper to make a permanent copy.



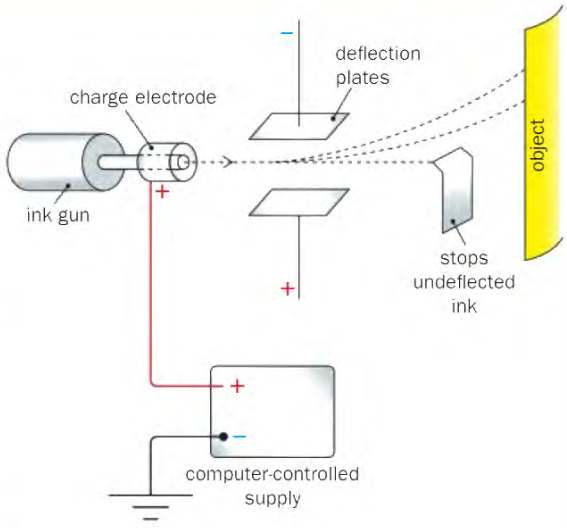
The ink-jet printer

One type of ink-jet printer is shown in the diagram:
The ink gun produces 10 000 tiny droplets of ink every second.
The droplets are positively charged by the charge electrode, which surrounds the ink gun nozzle.

The ink droplets are deflected by a vertical electric field.
In this printer, the p.d across the deflector plates is constant.

The voltage of the charge electrode is computer-controlled, so that each ink droplet is given a different size charge.
The greater the charge on the droplet, the more it is deflected as it passes between the deflector plates.
So the computer directs each droplet to a precise position.

How are the spaces between the words produced?
The charge electrode does not charge all the droplets.
Any uncharged droplets are not deflected.
They are stopped, as shown, and the ink is returned to the gun.
'Sell-by' dates are printed on cans using this technique.



Summary

The force F between two charges is given by Coulomb's law: $F = \dfrac{k Q_1 Q_2}{r^2}$ where $k = \dfrac{1}{4\pi\varepsilon_0}$

ε_0 is the permittivity of a vacuum or air.

The electric field strength E is the force acting per unit positive charge: $E = \dfrac{F}{Q}$

E is a vector quantity. It is measured in $N\,C^{-1}$ or $V\,m^{-1}$.

In a radial field: $E = \dfrac{k Q}{r^2}$ In the uniform field between two parallel plates: $E = \dfrac{V}{d}$

In a radial field: $V = \dfrac{k Q}{r}$ for a positive charge, and $V = -\dfrac{k Q}{r}$ for a negative charge.

The electric potential V gives the potential energy per coulomb of positive charge at a point in an electric field. It is defined as the work done per unit positive charge when the charge is moved from infinity to that point.

V is a scalar quantity. It is measured in volts. Energy transferred is $\Delta W = Q\,\Delta V$.

Questions

You *will need the following data:*
Permittivity of free space ε_0 is 8.85×10^{-12} F m^{-1}.
Charge on the electron is 1.6×10^{-19} C.
Mass of an electron is 9.1×10^{-31} kg.

1. Can you complete these questions?

a) The electric force between two charges depends on their .....and their .....

b) Charge is measured in .....
One ..... is the charge on 6.25×10^{18} .....

c) An electric field is a region in which a ..... feels a ..... Electric field lines can indicate the ..... and the ..... of a field.

d) In a uniform field the electric field lines are ..... and .....

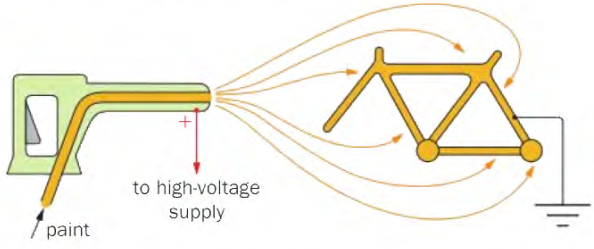
e) A surface joining all the points with the same potential is called an ..... surface.

f) To find the energy change when a charge moves from one point in a field to another, we multiply the ..... difference by the ..... on the object.

2. The diagram below shows an electrostatic paint spray.

a) Explain how the paint spray works.

b) Why does this technique waste little paint?



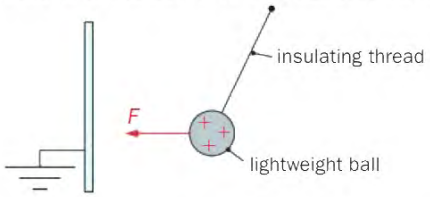
3. Describe two similarities and one difference between gravitational and electric fields.

4. Explain:

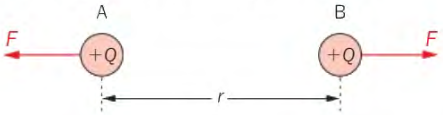
a) how two insulated conductors can be oppositely charged by induction,

b) why no work is done when a charge moves along an equipotential surface,

c) why the charged conducting ball (in the diagram below) is attracted to the earthed metal plate.



5. The two spheres A and B below are both given a charge $+Q$. The repulsive force between them is F.



What is the size of the repulsive force when:

a) the charge on A is doubled?

b) the charge on both spheres is doubled?

c) the charge on both the spheres is $+Q$, but the separation of the spheres is doubled to $2r$?

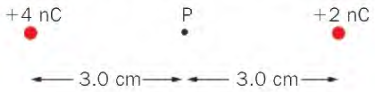
6. Calculate the electrostatic force between the proton and the electron in the hydrogen atom if the separation of the two particles is 1.0×10^{-10} m.

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7. Two small, equally charged spheres are 10 cm apart. The repulsive force between them is 3.6×10^{-6} N.
a) What is the charge on each sphere?
b) The spheres are negatively charged. How many excess electrons are there on each sphere?

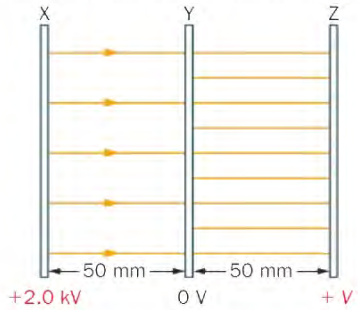
8. Calculate the electric field strength at distances of
a) 2.0 cm and b) 4.0 cm from a point charge of 16 nC. Comment on your answers.

9. Two point charges of +4.0 nC and +2.0 nC are placed 6.0 cm apart, as shown below.



a) What is the resultant field strength at point P, which is midway between the two charges?
b) What would be the field strength at P if the +2 nC charge was replaced with a −2.0 nC charge? (*Hint*: Find the field due to each charge and then add them together. Remember E is a vector.)

10. The diagram below shows three parallel, conducting plates, each connected to a different voltage.



a) What is the strength of the uniform electric field between plates X and Y?
b) What is the direction of the electric field between plates Y and Z?
c) What is the strength of the uniform electric field between plates Y and Z?
d) What is the potential V of plate Z?

11. A spark can pass through dry air when the electric field strength is 3.0×10^6 V m^{-1} or greater.
a) The gap between the electrodes in the spark plug of a car is 0.80 mm. What p.d. is needed across the electrodes to produce a spark?
b) An overhead electricity cable reaches a maximum voltage of 1.86×10^5 V. How close could you get to the cable before receiving an electric shock?

12. Phiz is foolishly standing in an open field during a thunderstorm. Draw the equipotentials between the cloud and the ground and use them to explain why Phiz is in danger of being struck by lightning.

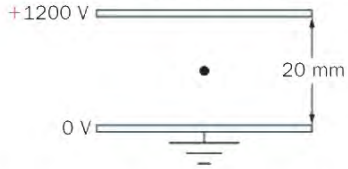
13. What is the force on a charge of 2.4 nC placed in an electric field of strength 4.0×10^3 N C^{-1}?

14. The diagram shows two metal plates, A and B. Plate B is earthed; plate A is at 1200 V.



a) What is the potential of each of the equipotential lines X, Y and Z?
b) A charge of -3.2×10^{-16} C is moved from X to Y. What is its change in electrical potential energy?
c) Has the system lost or gained electrical potential energy as the charge moves from X to Y?

15. A tiny negatively charged oil drop is held stationary in the electric field between two horizontal parallel plates, as shown below. Its mass is 4.0×10^{-15} kg.

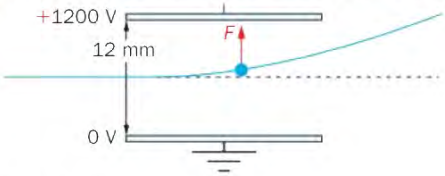


a) What are the two forces acting on the drop and in which direction do they act?
b) Use the fact that the 2 forces balance to calculate the charge on the oil drop. (g = 10 N kg^{-1}.)

16. The dome of a Van de Graaff generator has a radius of 15 cm.
a) What is the charge on the dome when the electric field strength at its surface is 3.0×10^6 V m^{-1}? (Remember, the sphere acts as if the charge is concentrated at its centre.)
b) What is the potential of the dome?

17. How much work is done when a charge of 3.6 pC is moved from 0.50 m to 0.75 m away from a point charge of −4.0 nC?

18. An electron beam passes between deflecting plates.



Calculate:
a) the electric field strength between the plates,
b) the deflecting force F on each electron.

Further questions on page 317.